Case Study: Employee Performance and Attrition Prediction

Introduction

In today's competitive business landscape, understanding the factors that influence employee performance and predicting employee attrition is crucial for organizations striving to improve their workforce management. Employee performance is a key determinant of an organization’s overall success, and high attrition rates can lead to increased recruitment and training costs, loss of productivity, and negative impacts on company morale. This project examines employee performance and attrition challenges at Sabor International Academy. By analyzing factors such as age, gender, department, job satisfaction, performance ratings, work-life balance, salary, tenure, job role, and overtime, the project aims to identify patterns associated with employee performance and attrition and support proactive retention planning.

Project Focus

The project uses workforce data to identify patterns associated with employee turnover and performance, engineer a composite performance index, and develop predictive classification models that can help identify employees who may require closer attention. The analysis is intended to provide a data-driven basis for earlier intervention and more targeted workforce retention decisions.

Project Scenario

Sabor International Academy is seeking to strengthen workforce management by using data analytics to understand employee performance and attrition patterns. Management has observed differences in job satisfaction, performance, and attrition across employees and departments. The organization wants to use workforce data to understand the underlying patterns and support proactive retention planning.

The data includes variables such as employee age, gender, department, job satisfaction, performance ratings, work-life balance, salary, years at the company, job role, overtime status, and whether an employee has left the organization (attrition). The goal is to identify patterns and predict which employees may be at higher risk of attrition, enabling earlier intervention and more targeted retention decisions.

Problems to be Addressed

1. **Identifying Key Factors Affecting Employee Performance:** Analyze patterns and relationships between performance ratings and factors such as job satisfaction, salary, age, and work-life balance.
2. **Predicting Employee Attrition:** Develop reliable classification models that can identify employees at higher risk of leaving, allowing HR and management to intervene early.
3. **Understanding the Relationship Between Job Satisfaction and Performance:** Investigate whether job satisfaction is associated with employee performance.
4. **Assessing the Impact of Overtime on Employee Performance and Attrition:** Explore whether overtime is associated with differences in performance or attrition.

Problems to be Addressed — Continued

5. **Evaluating Salary and Tenure Impact:** Examine how salary and years at the organization relate to employee retention and performance.
6. **Departmental Analysis:** Assess whether departments differ in performance and attrition patterns and whether factors such as job role and salary may help explain those differences.

List of Tasks to Solve the Problems

1. **Data Preprocessing:** Clean and prepare the data, including handling missing values and encoding categorical variables such as gender, department, job role, and overtime.
2. **Exploratory Data Analysis (EDA):** Explore the dataset using descriptive analysis and visualizations to identify patterns and relationships among employee factors, performance, and attrition.
3. **Feature Engineering:** Engineer a composite performance index using job satisfaction, work-life balance, and performance rating to represent employee performance across these dimensions.
4. **Model Development for Attrition Prediction:** Build and evaluate three classification models — **Logistic Regression, Decision Tree, and Random Forest** — to identify employees at higher risk of attrition.
5. **Analysis of Job Satisfaction vs. Performance:** Investigate the relationship between job satisfaction and performance ratings.

List of Tasks to Solve the Problems — Continued

6. **Overtime Analysis:** Explore how overtime relates to employee performance and attrition.
7. **Departmental Attrition and Performance Analysis:** Analyze department-level performance and attrition patterns to identify areas that may require focused attention.
8. **Model Evaluation:** Evaluate the three classification models using appropriate classification metrics and compare their performance for attrition prediction.

Conclusion

The analysis of employee performance and attrition data provides Sabor International Academy with a structured, data-driven approach to workforce management. By examining employee performance, job satisfaction, work-life balance, overtime, salary, tenure, and departmental patterns, the project supports a better understanding of factors associated with employee attrition.

The engineered performance index and three classification models provide a basis for identifying employees who may require closer attention. These insights can support earlier intervention, more targeted workforce retention decisions, improved employee satisfaction, and stronger workforce planning.

CV-Aligned Project Summary

Analyzed workforce data for Sabor International Academy to identify factors associated with employee attrition and support proactive retention planning; engineered a composite performance index across job satisfaction, work-life balance, and performance rating, and built and evaluated 3 classification models to identify employees at higher risk of attrition, providing a data-driven basis for earlier intervention and more targeted workforce retention decisions.